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Auto-antibodies to p53 and the subsequent development of colorectal cancer in a United States prospective cohort consortium

Julia Butt^{1,2}, William J. Blot³, Kala Visvanathan⁴, Loïc Le Marchand⁵, Lynne R. Wilkens⁵, Yu Chen⁶, Howard D. Sesso^{7,8}, Lauren Teras⁹, Marc D. Ryser¹⁰, Terry Hyslop^{2,11}, Sylvia Wassertheil-Smoller¹², Lesley F. Tinker¹³, John D. Potter^{14,15}, Mingyang Song^{8,16,17}, Sonja I. Berndt¹⁸, Tim Waterboer¹, Michael Pawlita¹, Meira Epplein²

¹Infections and Cancer Epidemiology, German Cancer Research Center (DKFZ), Im Neuenheimer Feld 280, 69120 Heidelberg, Germany ²Cancer Control and Population Sciences Program, Duke Cancer Institute, and Department of Population Health Sciences, Duke University, 2424 Erwin Road, Suite 602, Durham, NC 27705, USA ³Division of Epidemiology, Vanderbilt University Medical Center, 2525 West End Avenue Nashville, TN 37203, USA ⁴Department of Epidemiology, Johns Hopkins School of Public Health, 615 N. Wolfe Street, Baltimore, MD 21205 USA ⁵Epidemiology Program, University of Hawai'i Cancer Center, 701 Ilalo Street, Honolulu, HI 96813 USA ⁶Department of Population Health, New York University School of Medicine, 650 First Avenue, New York, NY 10016 USA ⁷Brigham and Women's Hospital, 75 Francis Street, Boston, MA 02115 USA ⁸Department of Epidemiology, Harvard T.H. Chan School of Public Health, 667 Huntington Avenue, Boston, MA 02115 USA ⁹Behavioral and Epidemiology Research Group, American Cancer Society, 250 Williams St, Atlanta, GA 30303, USA ¹⁰Department of Population Health Sciences, and Department of Mathematics, Duke University, 215 Morris St, Durham, NC 27701, USA ¹¹Department of Biostatistics and Bioinformatics, Duke University, 2424 Erwin Road, Durham, NC 27705 USA ¹²Department of Epidemiology & Population Health, Albert Einstein College of Medicine, 1300 Morris Park Avenue, Bronx, NY 10461 USA ¹³Cancer Prevention Program, Division of Public Health Sciences at Fred Hutchinson Cancer Research Center, 1100 Fairview Avenue, Seattle, WA 98109 USA ¹⁴Centre for Public Health Research, Massey University, Wellington, NEW ZEALAND 6140 and ¹⁵Fred Hutchinson Cancer Research Center, 1100 Fairview Avenue, Seattle WA 98109 USA ¹⁶Department of Nutrition, Harvard T.H. Chan School of Public Health, Boston, MA 02115 USA ¹⁷Clinical and Translational Epidemiology Unit and Division of Gastroenterology, Massachusetts General Hospital and Harvard Medical School, Boston, MA 02115 USA ¹⁸Infections and Immunoepidemiology Branch, Division of Cancer Epidemiology and Genetics, National Cancer Institute, National Institutes of Health, 9609 Medical Center Drive, Room SG/6E102, Rockville, MD 20850 USA

Abstract

Corresponding author: Julia Butt, PhD, Infections and Cancer Epidemiology, German Cancer Research Center (DKFZ), Im Neuenheimer Feld 280, 69120 Heidelberg, Germany, j.butt@dkfz-heidelberg.de.

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Background: Auto-antibodies to tumor suppressor p53 are found in a subset of colorectal cancer (CRC) patients. A recent prospective study in the US has reported a significant 1.8-fold increased odds for CRC development with pre-diagnostic sero-positivity to p53. In the present study, we sought to examine this association in a US CRC cohort consortium to evaluate the potential utility of p53 auto-antibodies as an early biomarker for CRC.

Methods: Auto-antibodies to p53 were measured in pre-diagnostic blood samples of 3,702 incident CRC cases and 3,702 controls, matched by age, race, and sex, from 9 US prospective cohorts. The association of sero-positivity to p53 with CRC risk, overall and by time between blood draw and diagnosis, was determined by conditional logistic regression.

Results: Overall, 5% of controls and 7% of cases were sero-positive to p53, resulting in a statistically significant 33% increased CRC risk (OR: 1.33; 95% CI: 1.09, 1.61). By follow-up time, the association was only significant with CRC diagnoses within 4 years after blood draw (OR: 2.27; 95% CI: 1.62, 3.19), but not thereafter (OR: 0.97; 95% CI: 0.76, 1.24).

Conclusions: In this large consortium of prospective cohorts, we found that pre-diagnostic sero-positivity to tumor suppressor p53 was significantly associated with an over 2-fold increased odds of developing CRC within 4 years after blood draw.

Impact: Our finding suggests that p53 sero-positivity may not be a useful predictor of long-term CRC risk, however, it might be considered as a marker to aid in the early diagnosis of CRC.

Keywords

Auto-antibodies; p53; serology; colorectal cancer; prospective

INTRODUCTION

Colorectal cancer (CRC) is the third most common cancer and the second most common cause of cancer death worldwide with more than 1.8 million newly diagnosed cases and 880,000 deaths in 2018 (1). Diagnosis of the cancer at an early stage has been shown to be associated with better survival (2), and screening by measurement of blood in stool as well as colonoscopy or sigmoidoscopy have been reported to be effective in reducing CRC mortality (3,4). However, although colonoscopy is the gold standard for the detection of CRC and precursor lesions, it is also a costly and invasive procedure (5). The less invasive blood-in-stool test is often impaired by the quality of the self-sampled specimen (6). Moreover, the current best performing assay for testing blood in stool (fecal immunochemical testing, FIT) still has limited sensitivity in detecting high-risk precancerous lesions, i.e. advanced adenomas (7,8). Thus, there is an ongoing search for new biomarkers that may identify individuals at risk of developing CRC, ideally in easily obtainable samples, such as blood.

It is well-known that a subset of CRC patients present with auto-antibodies directed against the tumor suppressor p53 (9). Missense mutations in the *TP53* gene lead to a loss of function of the protein and accumulation of the protein in the cancer cells to which 20–40% of patients develop antibodies (9,10). Various case-control studies have addressed the diagnostic performance of p53 auto-antibodies in detecting prevalent CRC and a recent

meta-analysis reported a pooled sensitivity of 19% and specificity of 93% (11). There is, however, only very little information available that addresses how long before diagnosis the association between p53 auto-antibodies and CRC can be observed and, thus, whether p53 serology might aid in identifying CRC patients at an early or even pre-malignant state. Pedersen et al., 2011, tested pre-diagnostic serial samples from 97 women who developed CRC and compared these to samples from 97 controls and found auto-antibodies to p53 peptides in 26% of the cases compared to 5% of controls up to 4 years prior to cancer diagnosis (12). A larger, prospective study by Teras et al., 2017, including 392 CRC cases and 784 controls reported that p53 sero-positivity was associated with an overall statistically significant 1.8-fold increased risk for a subsequent CRC diagnosis and a 2.3-fold increased risk when the analyzed blood sample was drawn within 3 years before CRC diagnosis (13).

In the present study, we sought to assess and potentially validate previous findings on the association of p53 auto-antibodies with the odds of developing CRC, by time from blood draw to diagnosis, in a consortium of 9 prospective cohort studies in the US.

MATERIALS AND METHODS

Study population

This study was conducted within a CRC cohort consortium designed as described previously (14). Briefly, the overall consortium included 4,063 prospectively ascertained CRC cases (C180–189, C100, and C209, International Classification of Diseases for Oncology, 3rd edition) and 1:1 matched controls nested within 10 U.S. cohort studies: Campaign Against Cancer and Stroke (CLUE), Cancer Prevention Study II (CPSII), Health Professionals Follow-up Study (HPFS), Multiethnic Cohort Study (MEC), Nurses' Health Study (NHS), NYU Women's Health Study (NYUWHS), Physician's Health Study (PHS), Prostate, Lung, Colorectal, and Ovarian Screening Study (PLCO), Southern Community Cohort Study (SCCS), and Women's Health Initiative (WHI). Controls were selected using incidence density sampling based on age at diagnosis and date of blood draw. Within each cohort, one control was chosen at random for each CRC case from the appropriate risk sets consisting of all cohort members who were alive and free of cancer (except non-melanoma skin cancer) at the time of diagnosis of the index case and had provided blood specimens. Matching criteria included sex, race, and date of birth. Because the association of p53 sero-positivity and CRC has already been assessed in the CPSII study (13), we excluded samples from CPSII for the current analysis, resulting in a total number of serum samples and socio-demographic data for 3,702 CRC cases and 3,702 controls. The median time between blood draw and diagnosis for cases was 7 years (range 0 to 40 years).

p53 serology

Antibody responses to p53 were measured as part of a larger panel of antigens, including bacterial and viral proteins, using multiplex serology as described previously (13–16). Briefly, full-length p53 (NCBI Accession number: NM_000546) was recombinantly expressed as Glutathione-S-transferase (GST)-tag fusion protein in *Escherichia coli* BL21 Rosetta (Novagen-Merck, Darmstadt, Germany) and affinity-purified on fluorescently labelled glutathione-coupled beads (Luminex Corp, Austin, TX, USA). Distinctly labelled

bead types, each carrying a different antigen, were mixed and incubated with serum (diluted 1:1000). Serum antibodies bound to the antigens were detected using a biotin-labelled secondary antibody (IgA/IgM/IgG) and subsequent incubation with the reporter fluorescent streptavidin-R-phycoerythrin. A Luminex 200 analyzer (Luminex Corp., Austin, TX, USA) then distinguished between the bead type and consequently the bound antigen as well as quantified the amount of bound serum antibody. Individuals with a median fluorescence intensity (MFI) of 200 or more against p53 were defined as p53 sero-positive (13). All personnel performing the serology assay were blinded for case-control status of the samples and all samples were analyzed in one batch to avoid potential batch-to-batch effects.

Statistical analysis

Differences in baseline socio-demographic factors between cases and controls, as well as between p53 sero-negative and -positive controls, were assessed using Chi-squared-test for categorical variables and Wilcoxon rank sum test for continuous variables.

Conditional logistic regression was used to determine odds ratios (OR) and 95% confidence intervals (CI) for the association of sero-positivity to p53 with the odds of developing CRC. The analysis was performed in the overall sample set, as well as by time between blood draw and diagnosis in 2 year increments. Based on these findings, we then divided time between blood draw and diagnosis in two categories (as <4 years or ≥4 years) to summarize the results. Apart from the matching variables age at blood draw, sex and race/ethnicity, we assessed the effect of *a priori*-determined potential confounders, including body mass index (BMI), smoking, education, a history of colonoscopy or sigmoidoscopy, and a family history of CRC. None of these variables changed the main OR by more than 10% and we therefore present the results from a conditional logistic model, with matched case-control sets as the strata, without further adjustment in the model.

In a sub-group analysis, we assessed the association of p53 sero-positivity with odds of developing CRC by cancer site, stage, and age at diagnosis in cases diagnosed within 4 years after blood draw and their matched controls using conditional logistic regression models. We also assessed whether the association varied by study, sex, and race/ethnicity by applying a multiplicative interaction term for the respective variable with p53 sero-positivity. Finally, we assessed the inclusion of data on a previous colonoscopy/sigmoidoscopy (available for n=807 controls and n=809 cases) on the association of p53 sero-positivity with CRC using an unconditional logistic regression model with adjustment for: age, sex, race/ethnicity; stratification by study; and the addition of a multiplicative interaction term.

All statistical analyses were performed using SAS 9.4 (SAS Institute, Cary, NC, USA).

RESULTS

Baseline characteristics

At baseline, CRC cases compared to controls were more frequently obese (BMI ≥30 kg/m²; 27% and 22%, respectively), more likely to have a family history of CRC (16% and 13%, respectively), and less likely to have undergone a colonoscopy or sigmoidoscopy prior to study recruitment (43% and 46%, respectively) (Table 1).

p53 sero-prevalence among controls varied by study, ranging from the highest in the SCCS (10%) to the lowest in WHI (2%). It was higher in males (7%) than females (4%) and in non-whites (8 – 10%) than whites (4%) (Table 2).

Association of p53 sero-positivity with CRC risk

Seven percent of all CRC cases (n = 242) were p53 sero-positive compared to 5% of controls (n = 185), resulting in a 33% increased odds of developing CRC in the overall consortium (OR: 1.33; 95% CI: 1.09, 1.61). Analysis of the association by time between blood draw and diagnosis showed that p53 sero-prevalence remained fairly stable in controls over time, varying between 3% and 6% (average 5%). In CRC cases, however, p53 sero-prevalence was highest (15%) in those with a CRC diagnosis within 2 years and lowest (3%) in those with a CRC diagnosis 14 or more years after blood draw (Figure 1A). The association of p53 sero-positivity with odds of developing CRC was statistically significant only in the sub-groups diagnosed within less than 2 years (OR: 2.73; 95% CI: 1.67, 4.45) and between 2 and 4 years (OR: 1.89; 95% CI: 1.17, 3.03) after blood draw (Figure 1B). Combining these two time-interval categories, the OR for developing CRC within 4 years after blood draw associated with p53 sero-positivity was 2.27 (95% CI: 1.62, 3.19). For any longer period between blood draw and diagnosis, there was no statistically significant association between p53 sero-positivity and odds of developing CRC (Figure 1B; OR for CRC diagnosis more than 4 years after blood draw: 0.97; 95% CI: 0.76, 1.24).

Analysis by case characteristics, including site, stage, and age at diagnosis, as well as by demographic factors of study, sex, and race, did not identify a specific group of CRC cases that was particularly strongly associated with baseline p53 sero-positivity (Table 3 and Supplementary Table S1). Moreover, stratification by screening history of colonoscopy or sigmoidoscopy had no statistically significant effect on the association of p53 sero-positivity and odds of developing CRC within 4 years after blood draw (no screening history: OR: 2.44; 95% CI: 1.46, 4.10; history of screening: OR: 2.08; 95% CI: 1.26, 3.43, $P_{\text{interaction}} = 0.496$) (Supplementary Table S1).

DISCUSSION

In this large CRC cohort consortium, we found that auto-antibodies to the tumor suppressor p53 were statistically significantly associated with an over 2-fold increased odds of developing CRC within 4 years after the analyzed blood sample was drawn. There was no association of p53 auto-antibodies with CRC when CRC was diagnosed 4 or more years after blood draw.

Our study is concordant with previously published studies to the extent that the observed association of p53 auto-antibodies with CRC is limited to a relatively short time period prior to CRC diagnosis. Pedersen et al., 2011, reported sero-conversion to p53 in longitudinal samples as early as 4 years and on average 2 years prior to CRC diagnosis (12). Teras et al. analyzed their prospective study in time intervals of <3, 3 to <6, and ≥6 years between blood draw and diagnosis. P53 sero-positivity was found to be statistically significantly associated with an increased 2.3-fold CRC risk only within the shortest time interval of <3 years (13). With our larger numbers, we were able to analyze our consortium data in shorter time

intervals (2-year increments) between blood draw and diagnosis. We replicated the findings of the two previous studies in that the association of p53 sero-positivity with CRC risk is particularly strong the shorter the interval between blood draw and diagnosis, with a 2.7-fold and 1.9-fold increased risk with a diagnosis within 2 years and between 2 and 4 years after blood draw, respectively. Expression of mutated p53 is hypothesized to be a late event in colorectal carcinogenesis, most likely at the transition from adenoma to cancer (17–19), which would explain the absence of association of CRC risk with auto-antibodies to p53 in blood samples drawn with a longer time interval before diagnosis. This is further supported by a study that assessed p53 sero-prevalence in non-advanced and advanced adenoma cases and did not find associations (20). Thus, p53 serology is unlikely to be applied as a biomarker for pre-cancerous lesions.

The potential diagnostic performance of p53 auto-antibodies in detecting CRC is further limited due to the low prevalence in CRC cases. A recent meta-analysis reported a pooled sero-prevalence of 19% among cases and 7% among controls in 11 case-control studies (11). The p53 sero-prevalence in our study is in line with these findings with 15% within 2 years prior to diagnosis and only 10% for 2 to 4 years prior to diagnosis, while on average 5% of controls were sero-positive. Teras et al., using the same methodology to measure auto-antibodies to p53, reported a similar sero-prevalence of 13% in cases that were diagnosed within 3 years after blood draw (13). Pedersen et al., however, analyzed their study using an array of p53 15-mer peptides and reported a sero-prevalence of 26% among cases and 5% among controls with the best performing peptides (12). Thus, modification of the p53 antigen might lead to higher sero-prevalences among cases. Previous literature has also discussed the inclusion of p53 into a panel of auto-antigens. Werner et al., 2016, for example combined p53 with carcinoembryonic antigen, osteopontin, seprase, and ferritin and detected 50% of colonoscopy-confirmed CRC cases compared to 10% of confirmed healthy controls. (21). A systematic review by Chen et al., 2014, showed marker panels including p53 that detected up to 88% of CRC cases; however, as the authors concluded, these panels warrant further validation in screening settings to assess non-inferiority to gold-standard screening (22). Additionally, auto-antibodies to p53 have been found with other cancers besides those of the colorectum, for example in individuals with breast, lung, and gastric cancer, as well as ovarian cancer and others (9); incorporating other auto-antigens could increase specificity to detect certain cancers.

A limitation of our study is the lack of knowledge of the underlying reasons for auto-antibodies to p53 in our control population. We observed differences in sero-prevalence varying between 2% and 10% by sex, race/ethnicity, and concordantly by study, which could suggest prevalent cancers in our control population, biasing the OR towards the null. The highest sero-prevalence was observed in the SCCS (10%), HPFS (9%), and MEC (9%). MEC and HPFS did not exclude controls who had a cancer diagnosis (except CRC) after diagnosis of the index case. As pointed out above, other cancers have been described to be associated with auto-antibodies to p53 (9), which could explain the presence of p53 sero-positivity in a proportion of controls. However, for example, in the MEC, only 6 of the 67 p53 sero-positive controls were linked to a cancer diagnosis after diagnosis of the index case. This does not explain the substantially higher number of p53 sero-positives compared to other studies. Another explanation could be differences in the level of immune response

by sex and/or race. It has been shown that sex and race influence the immune response elicited by an individual (23), which could also relate to differences in the adaptive immune response against auto-antigens. This hypothesis needs to be further elucidated and should be considered when attempting to implement antibody markers in cancer screening. As mentioned above controls selected by the majority of cohorts in this consortium potentially developed cancer after diagnosis of the index case. In four cohorts, however, (WHI, PHS, NYUWHS, and SCCS) controls were selected as super-controls, in that they did not develop cancer within the follow-up of the study. This, in turn, may introduce a bias that leads to an overestimation of our results. A sensitivity analysis excluding these cohorts showed that the results did not differ: overall 8% of cases were p53 sero-positive compared to 6% of controls (OR: 1.38; 95% CI: 1.10, 1.74); within four years after blood draw 15% of cases were sero-positive versus 7% controls (OR: 2.20; 95% CI: 1.51, 3.20). This suggests that selection of super-controls in some of the cohorts did not overestimate our findings. Major strengths of our study, however, are the large sample size; data on stage, tumor site, and age at diagnosis for the majority of cases; and evidence that the association was present across all categories of case characteristics. We moreover showed that a prior colonoscopy or sigmoidoscopy did not modify the association of p53 sero-positivity with odds of developing CRC. However, we lacked information on the time since the last CRC screening for the majority of study participants and were thus unable to perform a more detailed analysis of screening.

In conclusion, this study provides evidence that auto-antibodies directed against p53 are associated with increased odds of developing CRC but only for a subset of cancer cases and within a short time period before diagnosis. Thus, auto-antibodies to p53 may not be a useful predictor of long-term CRC risk, but could be considered within a marker panel for the early detection of CRC.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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REFERENCES

1. Ferlay J, Ervik M, Lam F, Colombet M, Mery L, Piñeros M, et al. 2018 Global Cancer Observatory: Cancer Today. Lyon, France: International Agency for Research on Cancer <<https://gco.iarc.fr/today>>.
2. Siegel RL, Miller KD, Fedewa SA, Ahnen DJ, Meester RGS, Barzi A, et al. Colorectal cancer statistics, 2017. *CA Cancer J Clin* 2017;67(3):177–93 doi 10.3322/caac.21395. [PubMed: 28248415]

3. Brenner H, Stock C, Hoffmeister M. Effect of screening sigmoidoscopy and screening colonoscopy on colorectal cancer incidence and mortality: systematic review and meta-analysis of randomised controlled trials and observational studies. *BMJ* 2014;348:g2467 doi 10.1136/bmj.g2467. [PubMed: 24922745]
4. Hewitson P, Glasziou P, Watson E, Towler B, Irwig L. Cochrane systematic review of colorectal cancer screening using the fecal occult blood test (hemoccult): an update. *Am J Gastroenterol* 2008;103(6):1541–9 doi 10.1111/j.1572-0241.2008.01875.x. [PubMed: 18479499]
5. Pawa N, Arulampalam T, Norton JD. Screening for colorectal cancer: established and emerging modalities. *Nat Rev Gastroenterol Hepatol* 2011;8(12):711–22 doi 10.1038/nrgastro.2011.205. [PubMed: 22045159]
6. Gies A, Gruner LF, Schrotz-King P, Brenner H. Effect of Imperfect Compliance With Instructions for Fecal Sample Collection on Diagnostic Performance of 9 Fecal Immunochemical Tests. *Clin Gastroenterol Hepatol* 2019;17(9):1829–39 e4 doi 10.1016/j.cgh.2019.03.001. [PubMed: 30851476]
7. Brenner H, Tao S. Superior diagnostic performance of faecal immunochemical tests for haemoglobin in a head-to-head comparison with guaiac based faecal occult blood test among 2235 participants of screening colonoscopy. *Eur J Cancer* 2013;49(14):3049–54 doi 10.1016/j.ejca.2013.04.023. [PubMed: 23706981]
8. Wong MC, Ching JY, Chan VC, Lam TY, Shum JP, Luk AK, et al. Diagnostic Accuracy of a Qualitative Fecal Immunochemical Test Varies With Location of Neoplasia But Not Number of Specimens. *Clin Gastroenterol Hepatol* 2015;13(8):1472–9 doi 10.1016/j.cgh.2015.02.021. [PubMed: 25724708]
9. Soussi T p53 Antibodies in the sera of patients with various types of cancer: a review. *Cancer Res* 2000;60(7):1777–88. [PubMed: 10766157]
10. Brosh R, Rotter V. When mutants gain new powers: news from the mutant p53 field. *Nat Rev Cancer* 2009;9(10):701–13 doi 10.1038/nrc2693. [PubMed: 19693097]
11. Meng R, Wang Y, He L, He Y, Du Z. Potential diagnostic value of serum p53 antibody for detecting colorectal cancer: A meta-analysis. *Oncol Lett* 2018;15(4):5489–96 doi 10.3892/ol.2018.8070. [PubMed: 29556296]
12. Pedersen JW, Gentry-Maharaj A, Fourkala EO, Dawney A, Burnell M, Zaikin A, et al. Early detection of cancer in the general population: a blinded case-control study of p53 autoantibodies in colorectal cancer. *Br J Cancer* 2013;108(1):107–14 doi 10.1038/bjc.2012.517. [PubMed: 23169294]
13. Teras LR, Gapstur SM, Maliniak ML, Jacobs EJ, Gansler T, Michel A, et al. Prediagnostic Antibodies to Serum p53 and Subsequent Colorectal Cancer. *Cancer Epidemiol Biomarkers Prev* 2018;27(2):219–23 doi 10.1158/1055-9965.EPI-17-0407. [PubMed: 29254936]
14. Butt J, Varga MG, Blot WJ, Teras L, Visvanathan K, Le Marchand L, et al. Serologic Response to *Helicobacter pylori* Proteins Associated With Risk of Colorectal Cancer Among Diverse Populations in the United States. *Gastroenterology* 2019;156(1):175–86 e2 doi 10.1053/j.gastro.2018.09.054. [PubMed: 30296434]
15. Butt J, Blot WJ, Teras LR, Visvanathan K, Le Marchand L, Haiman CA, et al. Antibody Responses to *Streptococcus Gallolyticus* Subspecies *Gallolyticus* Proteins in a Large Prospective Colorectal Cancer Cohort Consortium. *Cancer Epidemiol Biomarkers Prev* 2018;27(10):1186–94 doi 10.1158/1055-9965.EPI-18-0249. [PubMed: 30038049]
16. Waterboer T, Sehr P, Michael KM, Franceschi S, Nieland JD, Joos TO, et al. Multiplex human papillomavirus serology based on in situ-purified glutathione s-transferase fusion proteins. *Clin Chem* 2005;51(10):1845–53 doi 10.1373/clinchem.2005.052381. [PubMed: 16099939]
17. Baker SJ, Preisinger AC, Jessup JM, Paraskeva C, Markowitz S, Willson JK, et al. p53 gene mutations occur in combination with 17p allelic deletions as late events in colorectal tumorigenesis. *Cancer Res* 1990;50(23):7717–22. [PubMed: 2253215]
18. Laurent-Puig P, Blons H, Cugnenc PH. Sequence of molecular genetic events in colorectal tumorigenesis. *Eur J Cancer Prev* 1999;8 Suppl 1:S39–47. [PubMed: 10772417]
19. Srivastava S, Verma M, Henson DE. Biomarkers for early detection of colon cancer. *Clin Cancer Res* 2001;7(5):1118–26. [PubMed: 11350874]

20. Chen H, Werner S, Butt J, Zornig I, Knebel P, Michel A, et al. Prospective evaluation of 64 serum autoantibodies as biomarkers for early detection of colorectal cancer in a true screening setting. *Oncotarget* 2016;7(13):16420–32 doi 10.18632/oncotarget.7500. [PubMed: 26909861]
21. Werner S, Krause F, Rolny V, Strobl M, Morgenstern D, Datz C, et al. Evaluation of a 5-Marker Blood Test for Colorectal Cancer Early Detection in a Colorectal Cancer Screening Setting. *Clin Cancer Res* 2016;22(7):1725–33 doi 10.1158/1078-0432.CCR-15-1268. [PubMed: 26561557]
22. Chen H, Werner S, Tao S, Zornig I, Brenner H. Blood autoantibodies against tumor-associated antigens as biomarkers in early detection of colorectal cancer. *Cancer Lett* 2014;346(2):178–87 doi 10.1016/j.canlet.2014.01.007. [PubMed: 24462820]
23. Brodin P, Davis MM. Human immune system variation. *Nat Rev Immunol* 2017;17(1):21–9 doi 10.1038/nri.2016.125. [PubMed: 27916977]

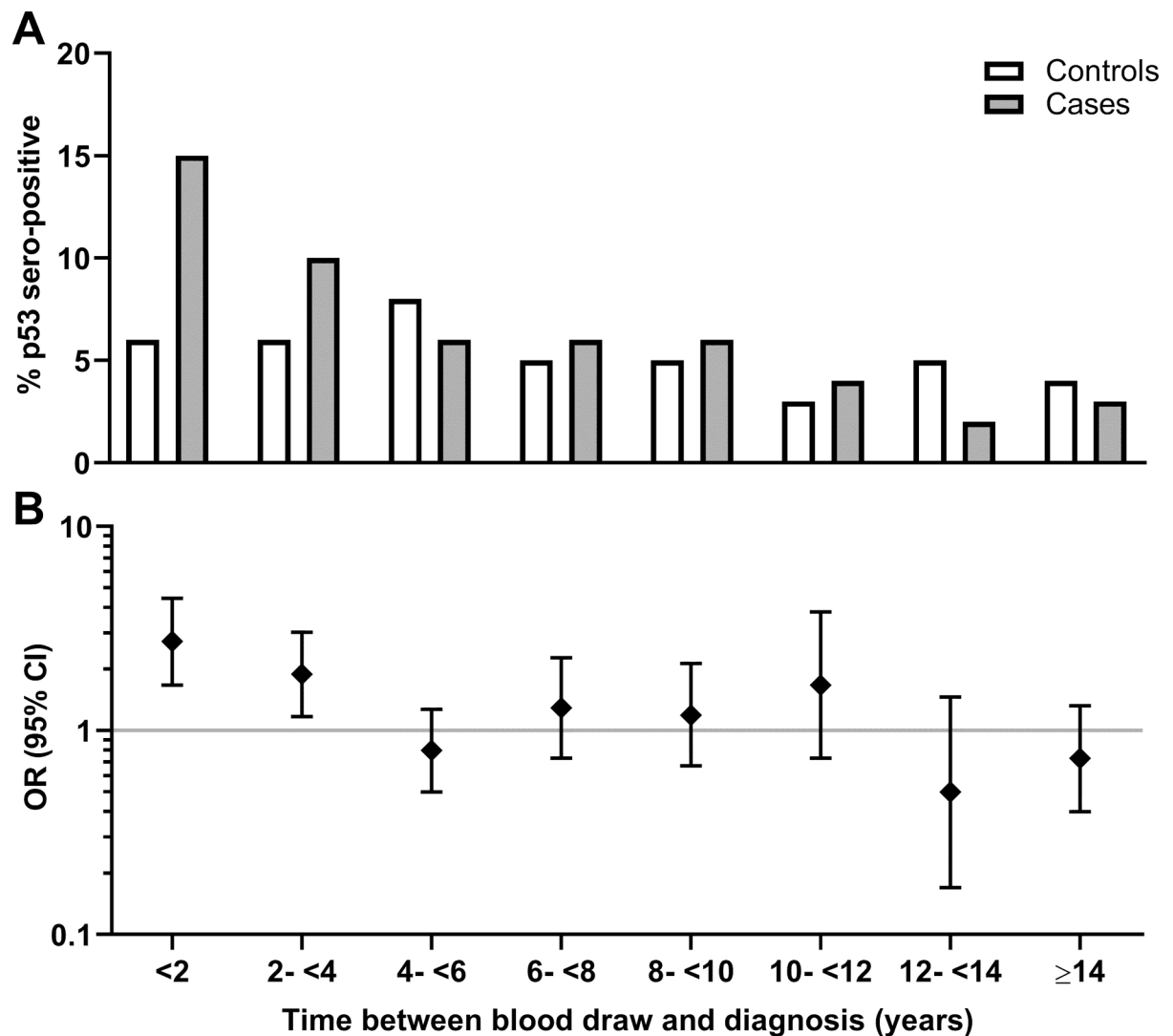


Figure 1: P53 sero-prevalence and association with CRC risk by time interval between blood draw and diagnosis.

Panel A) shows the p53 sero-prevalence in CRC cases and controls and panel B) the association of p53-sero-positivity with CRC risk, both in 2-year increments of time interval between blood draw and diagnosis. Controls are matched to cases by study, age, sex, and race/ethnicity. OR (diamond) and 95% CI (whiskers) in (B) were estimated by conditional logistic regression without further adjustment, the horizontal line at an OR of 1 serves as a reference for null association.

Table 1:

Sociodemographic factors in CRC cases and controls

Variable	Total (n=7,404)	Controls (n=3,702)	Cases (n=3,702)
Study, n (%)			
CLUE	982 (13)	491 (13)	491 (13)
HPFS	302 (4)	151 (4)	151 (4)
MEC	1,510 (20)	755 (20)	755 (20)
NHS	576 (8)	288 (8)	288 (8)
NYUWHS	572 (8)	286 (8)	286 (8)
PHS	360 (5)	180 (5)	180 (5)
PLCO	1,240 (17)	620 (17)	620 (17)
SCCS	252 (3)	126 (3)	126 (3)
WHI	1,610 (22)	805 (22)	805 (22)
Age at blood draw, years			
Median (range)	63 (18, 89)	63 (18, 88)	64 (18, 89)
Sex, n (%)			
Female	4,744 (64)	2,372 (64)	2,372 (64)
Male	2,660 (36)	1,330 (36)	1,330 (36)
Race/Ethnicity, n (%)			
White	5,422 (73)	2711 (73)	2711 (73)
African-American	792 (11)	396 (11)	396 (11)
Asian-American	612 (8)	306 (8)	306 (8)
Latino	422 (6)	211 (6)	211 (6)
Other/unknown/multiracial	156 (2)	78 (2)	78 (2)
Education, n (%)			
Less than HS	950 (13)	458 (13)	492 (14)
Completed HS or GED	1,515 (21)	750 (20)	765 (21)
Post HS training other than college	304 (4)	157 (4)	147 (4)
Some college	1,492 (20)	748 (20)	744 (20)
College graduate	1,330 (18)	682 (19)	648 (18)
Graduate school	1,707 (23)	865 (24)	842 (23)
BMI [kg/m²], n (%)			
<30	4,770 (75)	2,470 (78)	2,300 (73)
≥30	1,564 (25)	701 (22)	863 (27)
Smoking, n (%)			
Never	3,303 (45)	1,685 (46)	1,618 (44)
Former	2,983 (41)	1,439 (39)	1,544 (42)
Current	1,030 (14)	536 (15)	484 (14)
Ever had colonoscopy/sigmoidoscopy, n (%)			
No	3,003 (56)	1,453 (54)	1,550 (57)
Yes	2,404 (44)	1,252 (46)	1,152 (43)

Variable	Total (n=7,404)	Controls (n=3,702)	Cases (n=3,702)
Family history of CRC, n (%)			
No	4,558 (85)	2,329 (87)	2,229 (84)
yes	794 (15)	356 (13)	438 (16)

Abbreviations: HS, high school; GED, general educational development test

Table 2:

Variables associated with p53 sero-positivity among controls

Variable	Total n	P53 negative	P53 positive
Study, n (%)			
CLUE	491	477 (97)	14 (3)
HPFS	151	138 (91)	13 (9)
MEC	755	688 (91)	67 (9)
NHS	288	273 (95)	15 (5)
NYUWHS	286	274 (96)	12 (4)
PHS	180	171 (95)	9 (5)
PLCO	620	597 (96)	23 (4)
SCCS	126	114 (90)	12 (10)
WHI	805	785 (98)	20 (2)
Age at blood draw, years			
Median (range)	3,702	63 (18, 88)	65 (29, 87)
Sex, n (%)			
Female	2,372	2,281 (96)	91 (4)
Male	1,330	1,236 (93)	94 (7)
Race/Ethnicity, n (%)			
White	2,711	2,615 (96)	96 (4)
African-American	396	362 (91)	34 (9)
Asian-American	306	274 (90)	32 (10)
Latino	211	195 (92)	16 (8)
Other/unknown/multiracial	78	71 (91)	7 (9)
Education, n (%)			
Less than HS	458	438 (96)	20 (4)
Completed HS or GED	750	712 (95)	38 (5)
Post HS training other than college	157	151 (96)	6 (4)
Some college	748	711 (95)	37 (5)
College graduate	682	640 (94)	42 (6)
Graduate school	865	824 (95)	41 (5)
BMI [kg/m²], n (%)			
<30	2,470	2,335 (95)	135 (5)
≥30	701	667 (95)	34 (5)
Smoking, n (%)			
Never	1,685	1,607 (95)	78 (5)
Former	1,439	1,363 (95)	76 (5)
Current	536	507 (95)	29 (5)
History of colonoscopy/sigmoidoscopy, n (%)			
No	1,453	1,385 (95)	68 (5)
Yes	1,252	1,172 (94)	80 (6)
Family history of CRC, n (%)			

Variable	Total n	P53 negative	P53 positive
No	2,329	2,211 (95)	118 (5)
Yes	356	333 (94)	23 (6)

Abbreviations: HS, high school; GED, general educational development test

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Table 3:

P53 sero-positivity association with risk of developing CRC within less than 4 years after blood draw, by case characteristics

	N controls/cases	p53 positive controls, n (%)	p53 positive cases, n (%)	OR*	95% CI
Diagnosed within <4 years after blood draw	944/944	55 (6)	116 (12)	2.27	1.62, 3.19
By site					
Right colon	443/443	21 (5)	46 (10)	2.47	1.41, 4.34
Left colon	268/268	19 (7)	34 (13)	1.83	1.03, 3.26
Rectum	157/157	11 (7)	28 (18)	2.89	1.35, 6.17
By stage					
I/II	395/395	22 (6)	44 (11)	2.05	1.22, 3.45
III/IV	274/274	23 (8)	41 (15)	1.90	1.11, 3.27
By age at diagnosis					
≤65	339/339	19 (6)	46 (14)	2.69	1.51, 4.77
66 – 75	391/391	23 (6)	35 (9)	1.57	0.91, 2.72
>75	214/214	13 (6)	35 (16)	3.00	1.52, 5.94

* Conditional logistic regression, adjusted for the matching factors only.